Customer Journey Map

Date:	21 October 2026	**Team ID:**	Piyush-Balodiya-GenAI-2026
Project Name:	Multimodal Construction Intelligence (CONSTRUCTAI)	**Maximum Marks:**	2 Marks

Customer Journey Map (Part 1 - Stages 1 to 3)

Map out the customer's experience stage-by-stage, capturing their actions, touchpoints, thoughts, and feelings across the primary operational phases of the project.

Phase of Journey	Stage 1: Pre-Site Planning & Prep	Stage 2: Active On-Site Operations & Audits	Stage 3: Defect Resolution & Sign-Off
Actions What does the customer do?	Inputs project blueprint scope, material specifications, and cross-checks weather forecasts for planned concrete pours.	Uploads drone/CCTV feed snapshots to audit PPE compliance and captures wall/beam surface crack photos.	Reviews AI-generated IS 456 crack remediation reports, signs off on safety penalties, and exports executive audit logs.
Touchpoint What part of the service do they interact with?	CONSTRUCTAI Project Planning & Weather Hazard Advisory Portal.	Vision AI Safety Monitor & Structural Defect Analyzer module (Groq LLaMA 3.2 Vision).	Interactive Compliance Dashboard, Remediation Estimator & PDF Export Engine.
Customer Thought What is the customer thinking?	*“Will monsoon rain ruin this foundation batch? I need instant material BOQ quantities to prevent budget overrun.”*	*“Are workers wearing safety harnesses at height? Is that shear wall crack dangerous enough to stop construction?”*	*“I need verified IS-code standard repair methods quickly so management and clients approve work continuation.”*
Customer Feeling What is the customer feeling?	**Stressed & Rushed:** Under severe pressure to finalize site logistics and meet project milestones on time.	**Hyper-Vigilant:** Highly alert about fatal site accidents, falling hazards, and hidden structural defects.	**Relieved & Confident:** Reassured with structured repair blueprints, zero regulatory penalties, and audit transparency.

Customer Journey Map

Customer Journey Map (Part 2 - Process Ownership & Opportunities)

Mapping process leadership, accountability, and continuous improvement opportunities across each journey stage.

Phase of Journey	Stage 1: Pre-Site Planning & Prep	Stage 2: Active On-Site Operations & Audits	Stage 3: Defect Resolution & Sign-Off
Process Ownership **Who is in the lead on this?**	**Lead:** Project Planning Engineer **Support:** Procurement Lead & Structural Estimation Team	**Lead:** Site Safety Officer (EHS) **Support:** Lead Quality Control (QC) Inspector & Field Supervisors	**Lead:** Chief Project Manager **Support:** Structural Engineering Consultant & Client Auditors
Opportunities **How can we improve this stage?**	• **Automated Weather Sync:** Direct API integration with local Doppler radar to trigger real-time rain contingency alerts. • **BIM/CAD Auto-Parsing:** Instant material quantity takeoff (BOQ) parsing directly from uploaded construction blueprints. • **Supplier Price Integration:** Real-time material cost index matching with regional steel and cement vendors.	• **Edge-AI Deployment:** Sub-second low-latency inference on local site edge boxes for continuous CCTV streams. • **Multimodal Audio Sirens:** Automated loud speaker sirens alerting workers the second PPE gear is detected missing. • **Automated Severity Score:** Real-time classification of concrete cracks into micro, shear, or settlement hazards.	• **Automated ERP Procurement:** One-click purchase requisition for recommended epoxy resins and carbon fiber wraps. • **Digital Audit Trails:** Tamper-proof compliance certificate generation for local municipal authorities and insurance underwriters. • **Executive Analytics:** Weekly site hazard trends and contractor safety scoring dashboards.